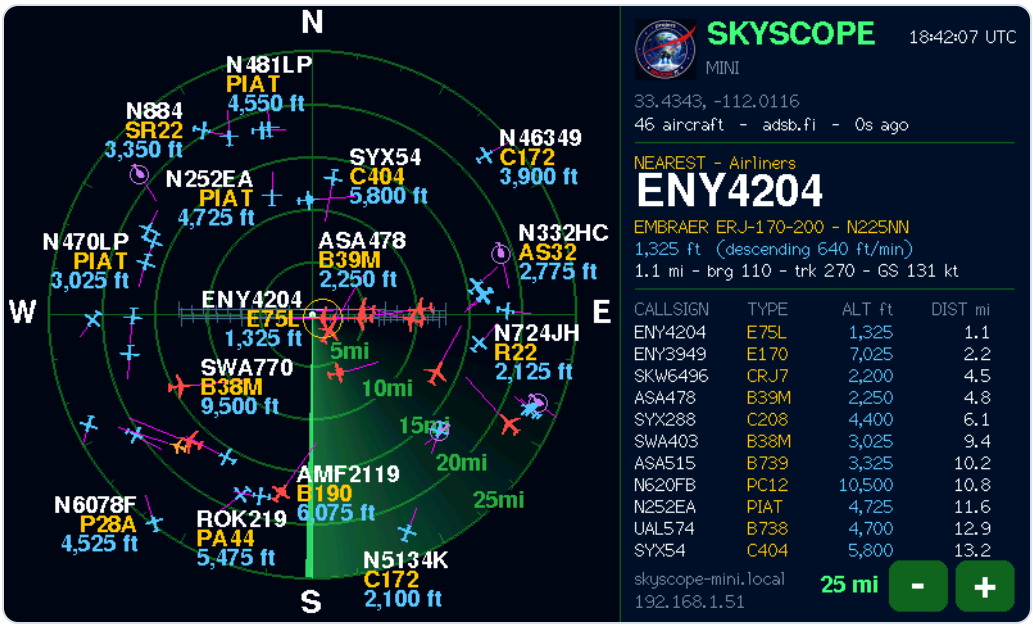
A PROJECT MOON PI PROJECT · USER GUIDE

SkyScope Mini

A small standalone aircraft radar on a 4.3-inch ESP32-S3 display. No Raspberry Pi needed.



SkyScope Mini centred on Phoenix Sky Harbor at 25 miles.

Version 1.1.1 · Project: github.com/ricsne2010/skyscope-mini · Licence: MIT · Not for navigation.

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1. What the Mini does

SkyScope Mini is a small, standalone aircraft radar on a 4.3-inch display. It connects to your Wi-Fi, fetches live aircraft positions from adsb.fi's open data every five seconds, and draws them on an air-traffic-control-style scope with a sweeping beam. No Raspberry Pi or other computer is needed once it's set up.

It's a reduced version of SkyScope, sized for the memory and speed of an ESP32 microcontroller:

The Mini has	Only on full SkyScope
Live aircraft from adsb.fi (up to 150 at once), smooth motion between updates	Your own SDR receiver; GPS location
Silhouettes or triangles, colour by kind, speed lines, tags, sweep	Airline logos and aircraft photos
Runways and extended centrelines at major airports	Smaller airports, airspace, SIDs and STARs
Filters by kind and altitude block; emergency squawk alerts	Type, operator and country filters; presets; watchlist
Ranges up to 100 km / 50 mi / 40 nm	Ranges up to 250 mi
Tap to select, –/+ range buttons, nearest-aircraft card, list, UTC clock	Browser radar, airspeed, full settings site

If you already run SkyScope on a Pi and want the full picture on a small screen, use SkyScope Satellite instead: it mirrors your SkyScope.

> **Early release.** Version 1.1.1 builds cleanly and its screen has been checked with the same drawing code running on a computer (see Building from source), but it has not yet been run on the board itself. If anything doesn't work, please open an issue with what you see.

2. What you need

Item	Details
Display board	**Sunton ESP32-8048S043**, sold as “4.3 inch ESP32-S3 display 800×480”; check the listing names the 8048S043. It has an ESP32-S3 with 16 MB of flash and 8 MB of PSRAM, an 800 × 480 screen and a touch panel. Choose the **capacitive** touch version (GT911) if offered; the resistive one works but its touch is not used.
USB cable	USB-C data cable (a charge-only cable won't let you install the firmware).
Power	Any 5 V USB supply of 1 A or more. Once installed, the computer is no longer needed.
Wi-Fi	A 2.4 GHz network (the ESP32 can't use 5 GHz).
Computer, once	To install the firmware: Chrome or Edge on Windows, macOS, Linux or ChromeOS (browser method), or PlatformIO (from source).
Internet	Through your Wi-Fi, to reach adsb.fi.
Case (optional)	A 3D-printable desk case styled like a radar console, in the case folder of the SkyScope Satellite project. It fits both the Satellite and the Mini; measure your board first, as its instructions explain.

3. Installing the firmware

There are two ways. Most people should use the browser: nothing to install on your computer.

From a web browser (recommended)

1. Download **skyscope-mini-1.1.1.bin** from the latest release (under *Assets*).
2. Plug the display into your computer with the USB-C cable.
3. Open https://espressif.github.io/esptool-js/ in **Chrome** or **Edge**. Leave the baud rate at 921600 and press **Connect**. Pick the port that appears (often “USB JTAG/serial debug unit” or “USB Single Serial”).

   > No port listed? Hold the board's **BOOT** button, press and release **RST** (or unplug and replug the cable), then let go of BOOT and press Connect again. On Windows a driver install is occasionally needed for the CH340 chip some boards use.

4. Under *Flash Address* enter `0x0`, choose the **skyscope-mini-1.1.1.bin** file, and press **Program**. It takes about a minute.
5. When it says it's finished, press the board's **RST** button (or unplug and replug it). The Project Moon Pi start-up screen appears.

Updating later is the same: program the newer file. Your Wi-Fi and settings are kept unless you choose *Erase Flash*.

From source with PlatformIO

For developers, or to change the code. Install PlatformIO (the VS Code extension or the command line), then:

```
git clone https://github.com/ricsne2010/skyscope-mini.git
cd skyscope-mini
pio run -t upload          # builds and installs over USB
pio device monitor         # optional: the board's log, 115200 baud
```

The first build downloads the ESP32 tools and libraries (LovyanGFX, WiFiManager, ArduinoJson) and takes a few minutes.

4. First start: joining Wi-Fi

1. After the start-up screen, a new display shows **Wi-Fi setup** instructions.
2. On your phone or laptop, join the Wi-Fi network `SkyScope-Mini-Setup` (no password).
3. A setup page opens by itself (if it doesn't, browse to `http://192.168.4.1`). Choose **Configure WiFi**, pick your network and type its password.
4. Press **Save**. The display joins your network and starts. It remembers the network from then on.

> To change networks later, or if the router changes, **hold the BOOT button for 3 seconds** while the Mini is running. It forgets the Wi-Fi and opens the setup network again. Keep holding to **10 seconds** instead to restore the default display wiring (see Display hardware).

5. Setting your location

The Mini starts centred on Amsterdam, so the first thing to do is tell it where you are.

1. On a phone or computer on the same Wi-Fi, open `http://skyscope-mini.local`. (If that doesn't open, use the numeric address shown at the bottom right of the Mini's screen.)
2. Under **Location**, enter your latitude and longitude in decimal degrees, for example `51.4700` and `-0.4543`. To find them, long-press your home on Google Maps or Apple Maps and copy the numbers; west and south are negative.
3. Choose the distance unit and range, then press **Save**. The screen follows straight away and fetches aircraft for the new area.

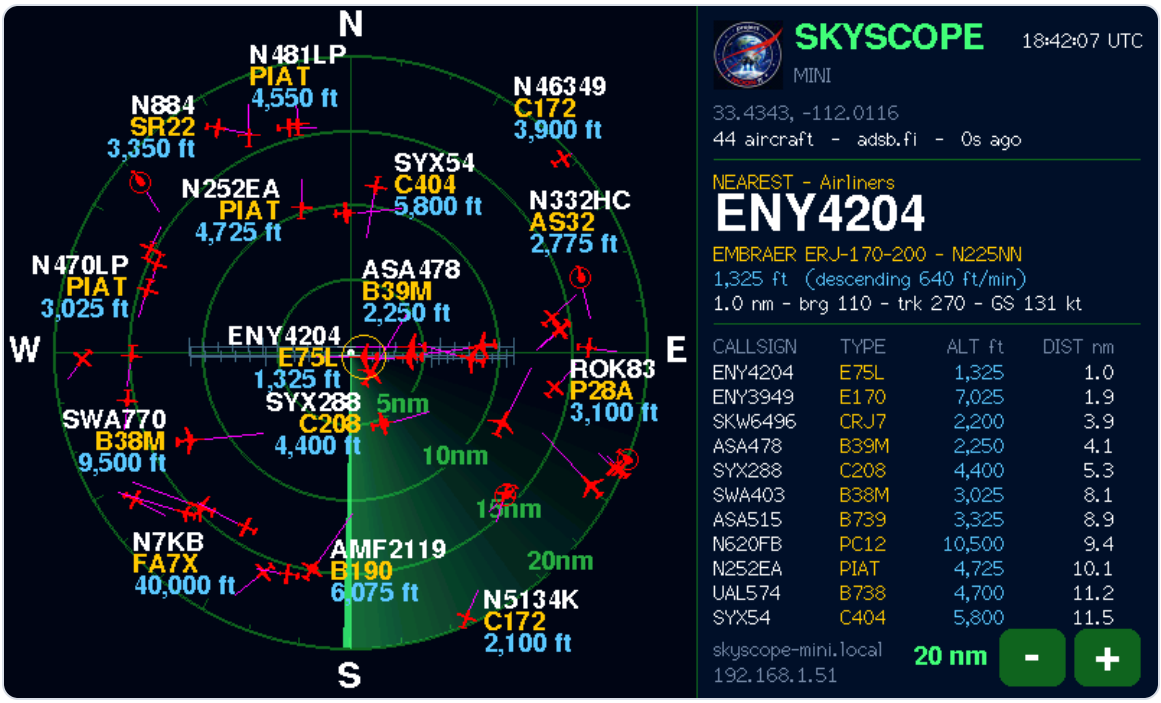
> Centring on an airport? Use the airport's position (its Wikipedia page lists it) and runways at major airports appear on the scope.

6. Reading the screen



Drawn by the Mini's code from live adsb.fi data: Phoenix Sky Harbor at 25 miles, colour by kind on, the runways with extended centrelines, the nearest aircraft on the card.

Part	Meaning
Rings	Distance from the centre, at round steps; each ring's distance is printed where it crosses the lower-right diagonal.
Symbols	Aircraft, pointing where they're heading, with a line showing about a minute of travel. Silhouettes show the kind: airliner, heavy, business jet, turboprop, light aircraft, glider, helicopter.
Colours	All red by default; with *Colour aircraft by kind*: airliners red, business jets orange, general aviation blue, helicopters purple, military khaki, others grey.
Tags	Callsign, type and altitude. When it's busy, tags that would overlap are left out (nearest aircraft first).
Yellow circle	The nearest aircraft, shown on the card at the right. A tapped aircraft gets a cyan circle and takes over the card.
Flashing red, pulsing ring	An emergency squawk (7500, 7600, 7700) or emergency status. Always shown, whatever the filters.
Card	Callsign, kind, aircraft type and registration, altitude and climb or descent, distance, bearing, track and ground speed.
List	Callsign, type, altitude and distance, nearest first, emergencies on top. It lists exactly the aircraft on the scope.
Top right	UTC time (set from the internet).



The same airspace at 20 nautical miles with colour by kind off.

7. Touch controls

Touch	Does
An aircraft	Selects it: cyan circle, and the card shows it until you tap it again, tap empty space, or it leaves the scope.
− / **+** (bottom right)	Next shorter or longer range, wrapping around. Saved straight away.

Unit	Ranges (to the outer ring)
Kilometres	5, 10, 15, 25, 50, 100
Statute miles	2, 5, 10, 15, 25, 50
Nautical miles	2, 5, 10, 20, 40

8. The settings page

At `http://skyscope-mini.local` (or the numeric address on the screen). Changes apply when you press **Save** and are kept when the power is off.

Section	Settings
Location	Latitude and longitude of the centre.
Display	Distance unit and range; silhouettes or triangles; colour by kind; sweep; major-airport runways with extended centrelines; aircraft on the ground; emergency highlighting.
Filters	Which kinds to show (airliners, business jets, general aviation, helicopters, military, other) and which altitude blocks (10,000 ft and below; 10,001 ft to FL180; FL180 to FL600).
About	Version, the Mini's address, the project link, credits, and a link to the Display hardware page.

The page has no password: anyone on your Wi-Fi can open it.

9. Display hardware (pins)

Boards sold as the ESP32-8048S043 have had small revisions, and similar boards use other pins. If the screen stays blank, the picture is shifted or rolls, the colours are wrong, or touch doesn't respond, you can change the display's wiring from a web page. You don't need to rebuild the firmware.

Open the **Display hardware** page in either of these ways:

- With the display on your Wi-Fi: `http://skyscope-mini.local/hardware`, or the link at the bottom of its settings page.
- With a blank screen, or if you don't know its address: hold **BOOT for 3 seconds** and let go. That starts Wi-Fi setup. Join `SkyScope-Mini-Setup` and press **Display hardware** on the setup page. (After saving the pins, choose *Configure WiFi* again to rejoin your network.)

Section	What's there
Panel data lines	The GPIO for each of the 16 colour bits: B0–B4, G0–G5, R0–R4. If red and blue come out swapped, swap the B and R rows.
Panel control	DE (HENABLE), VSYNC, HSYNC, PCLK and the backlight pin (-1 if there's none).
Panel timing	Pixel clock (4–20 MHz), the front and back porches and sync widths, the sync polarities, and the PCLK idle level. These come from the panel's data sheet. A picture that's shifted sideways or rolls usually needs different porches.
Touch	GT911 or none; SDA, SCL, reset and interrupt pins; and the address (0x14 or 0x5D).

Press **Save and restart**. The page checks the pins first: each must be a usable GPIO (1–21 or 38–48), and no pin can be used twice. Then the board restarts with the new wiring. **Restore defaults and restart** puts back the Sunton ESP32-8048S043 wiring, and so does holding **BOOT for 10 seconds**, even when you can't see the screen.

Default (ESP32-8048S043)	GPIO
B0–B4	8, 3, 46, 9, 1
G0–G5	5, 6, 7, 15, 16, 4
R0–R4	45, 48, 47, 21, 14
DE, VSYNC, HSYNC, PCLK	40, 41, 39, 42 (16 MHz; H porches 8/4/43, V 8/4/12; PCLK idle high)
Backlight	2
Touch (GT911)	SDA 19, SCL 20, reset 38, address 0x14

10. Troubleshooting

What you see	What to do
“Waiting for data…” for a long time	Check the Wi-Fi has internet access. adsb.fi is a free community service and is occasionally slow or down; the Mini keeps trying.
“adsb.fi answered 429”	Too many requests from your network (for example several Minis or other apps). It recovers by itself.
No aircraft	Check the location (west and south are negative), the filters, and try a longer range.
No runways	Only major airports are included, and only within the range plus about 25 km.
Touch does nothing	Your board may have resistive touch, or the touch chip may be on a different address or pins: try address 0x5D on the Display hardware page. The settings page does everything touch does.
Screen stays black, but the board's light is on	Press RST. If still black, your board revision may be wired differently: set its pins on the Display hardware page (it works without the screen). If you're stuck, open an issue with a photo of the board's back and its exact listing title.
Picture is shifted, flickers, or the colours are wrong (blue looks red)	Adjust the timing or pins on the Display hardware page. Red and blue swapped: swap the B and R rows.
Changed the pins and now nothing works	Hold BOOT for 10 seconds: the default wiring comes back and the board restarts. Or press **Restore defaults** on the Display hardware page.
“Not enough memory for the screen buffers”	The board has no PSRAM, so it is a different model. This firmware needs the 8 MB PSRAM version (ESP32-S3 N16R8).
Stuck on the Wi-Fi setup screen	Check the password, and that the network is 2.4 GHz. Join `SkyScope-Mini-Setup` again to retry.

Want to start over	Hold BOOT for 3 seconds and let go (forgets Wi-Fi), or use *Erase Flash* in the web installer and program the file again (forgets everything).

11. Building from source and the desktop preview

The code is in C++ for the Arduino framework, built with PlatformIO. The parts: `src/mini.cpp` saves settings and fetches from adsb.fi on its own task; `src/mini_core.cpp` holds ranges, rings, aircraft classification and the JSON reading; `src/render.cpp` draws the screen; `src/scope.cpp` holds the shared scope drawing (rings, symbols, tags, sweep); `src/main.cpp` starts everything, handles touch and serves the settings page. The major-airport runway table in `src/data/` is built from OurAirports by `tools/build_large_airports.py`.

`include/shapes.h` holds the aircraft silhouettes and type lists. It is generated from SkyScope's own `symbols.py` by `tools/gen_shapes.py`, so all the SkyScope projects draw and classify aircraft the same way.

Desktop preview

The drawing code has no hardware calls in it, so it also builds on a computer: `tools/preview` draws one frame from saved data into an image, exactly as the board would draw it. The pictures in this guide were made that way. It needs a C++ compiler and the SDL2 development files (LovyanGFX uses them to pick its desktop mode):

```
pio run                    # once, to download the libraries
tools/preview/build.sh     # SDL_INC=… SDL_LIB=… if SDL2 isn't installed system-wide
```

The comment at the top of `tools/preview/preview.cpp` shows its arguments.

12. Credits and licence

SkyScope Mini comes full circle: SkyScope grew out of ESP32-Plane-Radar by MatixYo (MIT licence), a radar for a small round ESP32 display, and the Mini brings SkyScope back to an ESP32. The runway table comes from that project's airport data tool.

What	Source	Terms
Live aircraft	adsb.fi open data	adsb.fi terms (at most one request a second; the Mini uses one every five)
Airports and runways	OurAirports	Public domain
Graphics	LovyanGFX	FreeBSD licence
Wi-Fi setup	WiFiManager	MIT
JSON	ArduinoJson	MIT

> **Not for navigation.** Aircraft positions may be delayed, incomplete or wrong. ADS-B shows only aircraft that broadcast it. Never use this display for flight planning or navigation.



SkyScope Mini is a Project Moon Pi project and free software under the MIT licence.

Source: github.com/ricsne2010/skyscope-mini.